Q1 – Unique Signup Simulation

Write a Python function based, user friendly program that:

- Creates user login with username & password and store it in a text file
- Allows user to choose a unique userID that is already not available in the text file.
- If login ID is successfully selected → Display welcome message
- While taking password, give retype password option also. And if both the passwords are equal than and then only sign up.

Q2 – Guess the 4 letter word

Develop a user friendly, function based program that:

- Accepts a four character word from the user. OR can take random characters from the list of alphabets

  [Hint: **import random**

  character_list = ['a', 'b', 'c', 'd', 'e', 'f', 'g',…….]

  random_char = random.choice(character_list)]

- Prints "Success" or "Try Again" based on the word is available in the file or not.
- Give three chance to guess the word from the file.

Q3. Smart Insurance Premium Calculator

An insurance company wants a Python program to calculate annual insurance premium based on risk category and policy rules.

1. Accept User Inputs

- Coverage Amount (Sum Assured)
- Policy Term (in Years)
- Applicant Age
- Plan Type

Plan Type options:

- Standard
- Premium

Early Validation Rules

Reject input if:

- Coverage amount ≤ 0
- Policy term ≤ 0
- Policy term > 40 → `"Policy term exceeds maximum allowed limit"`
- Age < 18 or Age > 70 → `"Applicant not eligible for policy"`
- **2. Apply Base Premium Rate (Based on Coverage Amount)**

Coverage Amount Range	Base Rate (% of Coverage per year)
Up to ₹5,00,000	2%
₹5,00,001 – ₹20,00,000	3.5%
Above ₹20,00,000	5%

3. Risk Loading Based on Age

Age Range	Extra Charge
18 – 30	No extra charge
31 – 50	+1% of Coverage
Above 50	+2% of Coverage

4. Plan Benefits

- If Plan Type is **Premium** → Add ₹2,000 fixed annual service charge
- If Policy Term > 20 years → Give 5% discount on total premium

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5. Premium Calculation Rules

Step 1:
Calculate Base Premium:

BasePremium = Coverage × (BaseRate/100)

Step 2:
Add Age Risk Charge (if applicable)

Step 3:
Add Premium Plan charge (if selected)

Step 4:
Apply Long-Term Discount (if eligible)

Final Annual Premium = Total Premium after adjustments

Monthly Premium = Final Annual Premium / 12

Display Output

Show:

- Applicable Base Rate (%)
- Age Risk Charge
- Plan Charge (if any)
- Discount (if applied)
- Final Annual Premium
- Monthly Premium

Q1 – Secure Login Simulation

Write a Python function based, user friendly program that:

- Stores predefined username & password in a text file
- Allows user **3 attempts** to login
- If login successful → Display welcome message
- Else → Block access with proper message

Q2 – Learn Mathematics

Develop a user friendly, function based program that:

- Accepts a number from user, make sure that number is <=10.
- Prints multiplication table up to that number.

Q3 – Smart Loan Calculator

A bank wants a Python program to calculate loan repayment details based on compound interest and risk policies.

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1. Accept User Inputs

- Loan Amount (Principal)
- Number of Years
- Customer Type

Customer Type options:

- Regular
- Senior Citizen

Early Validation Rules

Reject input if:

- Loan amount ≤ 0
- Years ≤ 0
- Years > 30 → "Loan tenure exceeds policy limit"

2. Apply Interest Rate Slabs

The bank assigns **annual interest rates based on loan amount**:

Loan Amount Range	Interest Rate
Up to ₹50,000	8%
₹50,001 – ₹2,00,000	10%
Above ₹2,00,000	12%

If customer is **Senior Citizen** → reduce rate by **1%**

3. Interest Calculation Rule

Use **Compound Interest Formula**: $A = P (1 + R/100)^T$

Where: P = Principal, R = Annual Interest Rate, T = Number of Years

Interest = A – P

EMI Formula: Monthly EMI = Total Amount / (Years × 12)

Display Output

Show:

- Applicable Interest Rate
- Total Interest
- Total Payable Amount
- Monthly EMI